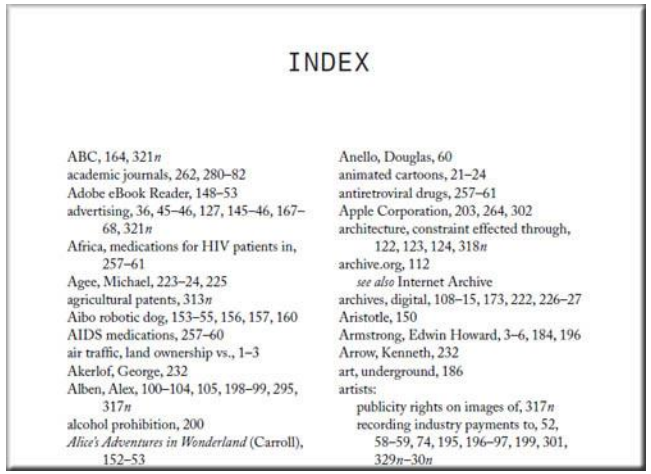
Rodzaje indeksów

```
CREATE INDEX index_name
ON table_name (column1, column2, ...);
```

- Wymaga miejsca
- Wymaga aktualizacji (balansowania struktury)
- Jest nadmiarowy
- Zawiera w sobie dane i wskaźnik jak te dane znaleźć



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https://www.postgresql.org/docs/12/indexes-types.html

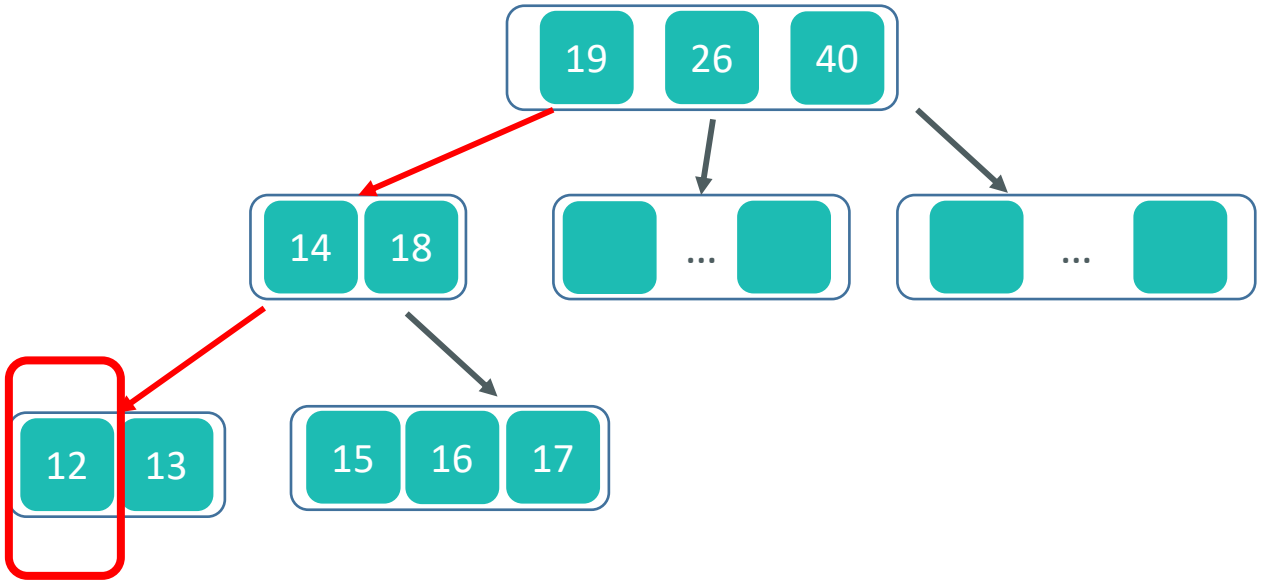
Rodzaje indeksów w PostgreSQL

CREATE INTEX idx_name ON table_name USING TYPE (COLUMN_NAME)

- **BTREE – Balanced Search Tree** – domyślny typ indeksu
- **HASH INDEX** – gdy korzystasz z operatora =
- **GiST** – dla operacji na mapach i operatorach GiS (<<, &<, >> ...)
- **SP-GiST –** podobnie jak GiST
- **GIN** – inverted index – dla wartości o wielu elementach (tablice / zbiory / JSON / wyszukiwania operatory <@, @>, = , &&)
- **BRIN (Block Range Indexes) –** podsumowania wartości przechowywanych w następujących po sobie fizycznych blokach tabeli

https://www.postgresql.org/docs/12/indexes.html

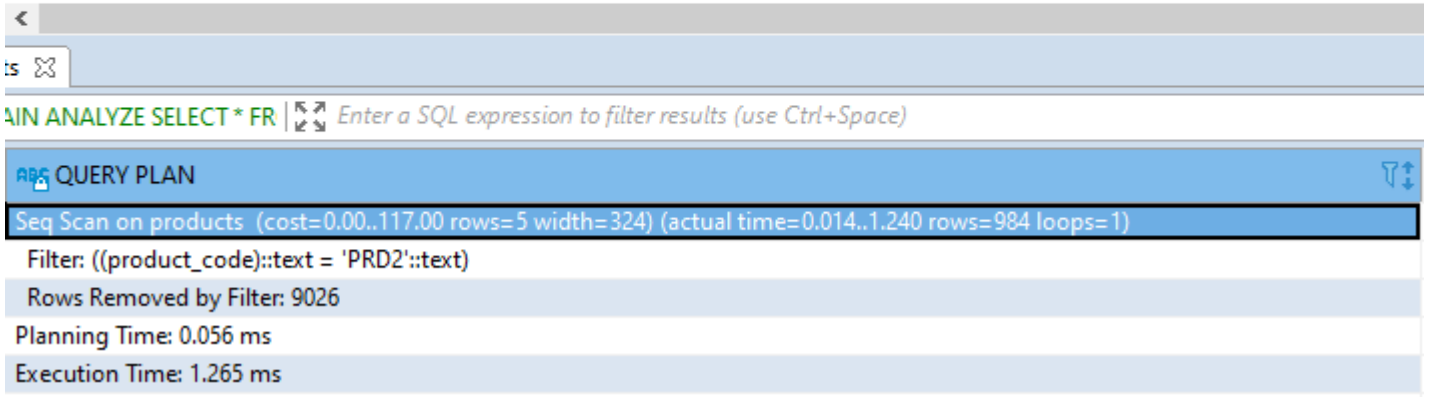
Balanced Search Tree (B-Tree)



Indeksy Wielokolumnowe

```
CREATE INDEX idx_products_name_code
    ON products
        USING BTREE (product_name,product_code);
```

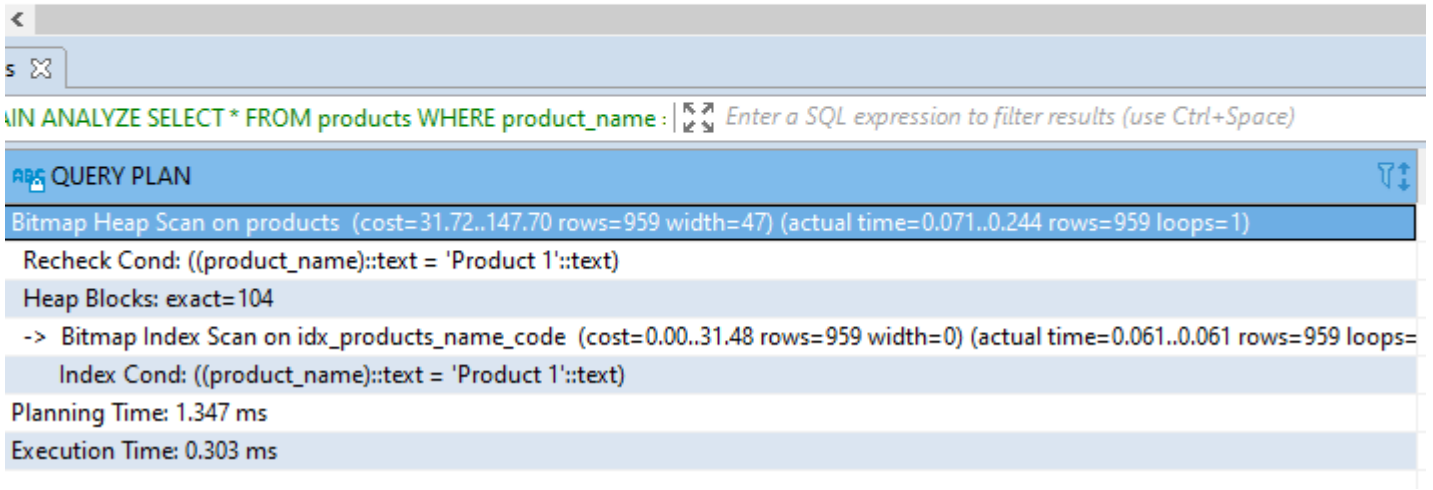
```
EXPLAIN ANALYZE
SELECT *
  FROM products
 WHERE product_code = 'PRD2';
```



Indeksy Wielokolumnowe

```
CREATE INDEX idx_products_name_code
    ON products
        USING BTREE (product_name,product_code);
```

```
EXPLAIN ANALYZE
SELECT *
  FROM products
 WHERE product_name = 'Product 1';
```



Indeksy Funkcyjne

```
CREATE INDEX idx_products_upper_name
    ON products
        USING BTREE (upper(product_name));
```

```
EXPLAIN ANALYZE
SELECT *
  FROM products
 WHERE upper(product_name) = 'Product 1';
```

